

Restock the Refrigerator SOP (1x Restock, in situ)

One episode restocks one refrigerator, in place.

The base is **passive**. It has no drive of its own. It is pushed by hand to the refrigerator and locked there. Nothing is carried away to a table. Everything the episode touches is already at the refrigerator when recording starts: the older stock inside it, and the seven new items standing in the stock tote on the stand in front of the base.

The refrigerator is worked **as found**. The door stands open on its catch when recording starts. The working shelf carries three groups of older stock. The crisper drawer under it holds one older produce pack. The two door bins each hold one older item. The episode ends with all seven new items away, every older item standing in front of the new stock behind it, the drawer shut, and the door pressed fully shut.

This is an in-situ task. Three rules follow from that.

First, there is a shelf directly above the working shelf. So **every approach to the shelf is from the front, straight in, level**. No gripper comes down onto the working shelf from above.

Second, **the refrigerator and the door are never leaned on and never pushed**. No gripper, wrist, or forearm rests on the door, the shelf edge, or the drawer face. No push is ever hard enough to move the appliance.

Third, **older stock is slid, never lifted**. It is drawn forward along the shelf, the drawer floor, or the bin floor. Nothing that starts inside the refrigerator comes out of it.

Every place is loaded the same way: rotate first, then fill. The older item in that place is drawn forward to the front, and the new item goes in behind it. New stock never goes in front of older stock, anywhere.

The order never changes. Load the crisper drawer. Load the three shelf groups. Load the two door bins. Close the door. The drawer is shut before anything goes on the shelf, so nothing is ever carried over an open drawer. The door bins are loaded last, so nothing is carried over the loaded shelf.

The two arms never cross. The door hangs open to the **left** on its own catch, so the **left gripper** works everything on the left: the carton group and both door bins. The **right gripper** works everything on the right: the drawer, the jar group, and the tub group. The stock tote sits in front of the base with a divider down it, and each gripper takes only from its own half. **Neither gripper ever holds the door during the loading**, and the left gripper always stays left of the right gripper.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the refrigerator. Stop it **square to the front of the appliance**, so the front edge of the working shelf runs straight across the frame and neither end of it sits nearer than the other.
2. Stop it **centered on the working shelf**, so the left end and the right end of the shelf are the same distance out from the middle of the base.
3. Stop it **close enough** that both grippers reach the back of the working shelf straight in through the door mouth without an arm extending. Stop it **far enough** that no arm, wrist, or part of the base touches the appliance or the open door while both arms work.
4. Check the **height band**. With the base parked, both grippers enter the door mouth level and reach the working shelf board from front to back. The **right gripper** reaches the crisper drawer handle and the whole drawer floor when the drawer is open. The **left gripper** reaches both door bins from above. None of this makes a wrist or forearm touch the shelf above.
5. Check the **left lane**. The **left gripper** reaches the carton group at the left end of the shelf, both door bins on the open door, and the left half of the stock tote, all without extending and without reaching right of the middle of the base.
6. Check the **right lane**. The **right gripper** reaches the drawer handle, the jar group, the tub group, and the right half of the stock tote, all without extending and without reaching left of the middle of the base.
7. Check the **door catch**. The door stands open on its catch on its own, clear of the whole door mouth and clear of both arms' paths, and it does not drift while the arms work.
8. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. Do not work a refrigerator the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the furniture hard enough to shift it. No operator touches it. It is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the door mouth. Its frame includes the whole working shelf, the crisper drawer shut and open, the open door with both bins, the door seal all round the mouth, and the stock tote on its stand.
3. The camera reads the working shelf from the front. You can see which group an item stands in, whether it stands upright, and whether anything sticks out past the shelf front edge.
4. The camera reads both door bins from above. You can see what is in each bin and where in the bin it stands.
5. The camera reads the door front from the front once it is shut. You can see any gap along the seal.
6. Both arms are at home with grippers open.
7. The **left arm** reaches the left half of the stock tote, the carton group at the left end of the shelf, and both door bins, without extending to a joint limit. It never needs to reach the drawer handle, the jar group, the tub group, or the right half of the tote.
8. The **right arm** reaches the right half of the stock tote, the drawer handle and the whole drawer floor, the jar group, and the tub group, without extending to a joint limit. It never needs to reach the carton group, a door bin, or the left half of the tote.
9. Each gripper enters and leaves the door mouth **from the front and level**. The **left gripper** comes onto each door bin straight down from above. Neither gripper comes down onto the working shelf from above, and neither wrist nor forearm touches the shelf above on the way in or out.
10. The two arms do not touch each other. The left gripper stays left of the right gripper throughout. The open door on its catch is clear of both arms' paths to the shelf, the drawer, and the tote.
11. If a place cannot be reached, re-park the base by the Base positioning steps until lines 7 to 10 hold.

Materials checklist

1. The **refrigerator** stands where it lives. It is not moved, not leaned on, and not pushed at any point.
2. The **door** is hinged on the **left** of the appliance. At the start of the episode it stands open to the left, held by a **door catch**, wide enough that the whole door mouth is clear and both bins face into the workspace on the left.
3. The **door catch** holds the door open on its own. The door does not swing in or drift out while the arms work, and it comes off the catch under one steady push without being unhooked by hand.

4. The door shuts with a latch or a magnet seal that holds it closed. A correctly shut door meets the cabinet all round, shows no gap along the seal, and does not spring back open.
5. The **working shelf** is the lowest shelf of the refrigerator, with a **shelf directly above it**. So the only way in is the front-facing door mouth, never the top.
6. There is clear space above the working shelf, so an item stands on it without touching the shelf above and a gripper enters level above the shelf board without touching that shelf.
7. The working shelf is dry and level. It carries **three groups of older stock**, standing upright, each group at the back of its own part of the shelf, none touching another: **cartons at the left end, jars in the middle, tubs at the right end**. Each group holds exactly one older item at the start.
8. The three kinds are visibly different from the front. A carton, a jar, and a tub are never a close call. Each group has clear shelf behind its older item for a second item of that kind.
9. Nothing else stands on the working shelf. Nothing on it is marked, taped, or labelled out.
10. The **crisper drawer** is the drawer directly below the working shelf. Its handle is at the middle of the drawer face, within the **right gripper's** reach. It pulls straight out toward the front to a stop, does not come free, and slides when drawn by its handle without being lifted.
11. **One older produce pack lies flat at the back of the crisper drawer**. The drawer is shut at the start.
12. The drawer floor is long enough front to back for three produce packs to lie flat in one row, and no wider than one pack. So there is one row and no choice of lane.
13. The door carries **two open topped bins side by side at the same height** on its inner face: the **tall bin** on the hinge side and the **short bin** on the free-edge side. Both are open above along their whole length, with clear space over each. Both are loaded straight down from above by the **left gripper**.
14. **One older bottle stands in the tall bin** and **one older small jar stands in the short bin**, each at the back of its bin against the door face. Both bins are otherwise empty.
15. Nothing in either bin stands high enough to touch the working shelf or the shelf above when the door swings shut.
16. The **stock stand** stands on the floor directly in front of the base, below and clear of the door mouth, at a height where both grippers lift an item straight up out of the tote without extending. It does not slide when an item is lifted.
17. The **stock tote** stands open on the stand, its mouth facing up, with a fixed **divider** down its middle splitting it into a **left half** and a **right half**. The divider stands high enough to read from the front.
18. The tote holds the **seven new items** standing upright side by side, none on top of another and none touching:
 - **left half**: one carton, one bottle, one small jar;
 - **right half**: one jar, one tub, two produce packs.

19. Each new item matches one older item by kind, so where it goes is never a judgment call: the carton to the carton group, the jar to the jar group, the tub to the tub group, the bottle to the tall bin, the small jar to the short bin, both produce packs to the drawer.
20. Every item in the tote's left half belongs to a place the **left gripper** works. Every item in its right half belongs to a place the **right gripper** works. No item ever crosses the divider.
21. Every new item stands on its own base unaided, is closed and dry, and is held by one gripper without its contents spilling.
22. Each produce pack is a sealed rigid pack that lies flat on one face, keeps its shape when gripped, and is short enough to lie in the drawer with clear space above it when the drawer is shut.
23. The floor and the space between the stand and the door mouth are clear, so an arm carrying an item passes over nothing.

Workspace layout

Nothing anywhere is marked or taped out. You judge every place below by eye against the appliance itself: the door mouth, the shelf front edge, and the stock already standing in place.

- **Refrigerator:** the appliance the base is parked at. It is never moved, never leaned on, and never pushed.
- **Door mouth:** the open front of the refrigerator. Both grippers go in and come out through it, level and from the front.
- **Working shelf:** the lowest shelf, and the only shelf worked. All shelf work happens on it.
- **Shelf above:** the shelf directly above the working shelf. It is what makes every shelf approach a front approach. Nothing ever touches it.
- **Shelf front edge:** the front edge of the working shelf board. Nothing may stick out past it, or the door will not shut.
- **Carton group:** the stretch of the working shelf at its **left end** where the older carton stands. **Left gripper only.**
- **Jar group:** the stretch in the **middle** where the older jar stands. **Right gripper only.**
- **Tub group:** the stretch at the **right end** where the older tub stands. **Right gripper only.**
- **Crisper drawer:** the drawer directly below the working shelf. **Right gripper only.**
- **Door:** the refrigerator door, hinged on the left, held open on its **door catch** for the whole of the loading. Neither gripper holds it while the stock goes in.
- **Door catch:** the stay that holds the door open on its own. The door comes off it under one steady push in Step 4.
- **Tall bin** and **short bin:** the two open topped bins on the door's inner face, the tall one on the hinge side and the short one on the free-edge side. With the door open to the left, both sit in the left lane. **Left gripper only.**
- **Stock stand:** the stand in front of the base holding the stock tote.

- **Stock tote:** the divided tote on the stand. Its **left half** holds the carton, the bottle, and the small jar. Its **right half** holds the jar, the tub, and the two produce packs. Empty at episode end.

Arm lanes

The two arms never cross. This holds for the whole episode.

- The **left gripper works the left lane:** the left half of the stock tote, the carton group, both door bins, and the push that takes the door off its catch.
- The **right gripper works the right lane:** the right half of the stock tote, the crisper drawer, the jar group, the tub group, and the final press that seals the door.
- The **left gripper always stays left of the right gripper.** Neither arm reaches over, under, around, or past the other. Neither arm reaches across the front of the other arm's body.
- **Neither gripper holds the door during the loading.** The door catch does that.
- Nothing is handed between grippers, and no item ever crosses the tote's divider. Only one item is moved at a time.

Arm assignments

- **Left gripper.** Takes the carton, the bottle, and the small jar from the tote's left half. Rotates and loads the carton group. Rotates and loads the tall bin and the short bin. Pushes the door off its catch and round to the door mouth in Step 4.
- **Right gripper.** Takes the jar, the tub, and the two produce packs from the tote's right half. Opens and shuts the crisper drawer and loads it. Rotates and loads the jar group and the tub group. Presses the door home at its free edge in Step 4.

Vocabulary

- **Older stock:** anything already inside the refrigerator when recording starts. That is the older carton, jar, tub, bottle, small jar, and produce pack. Older stock never comes out of the refrigerator.
- **New stock:** the seven items standing in the stock tote when recording starts. All seven go in during the episode.
- **Rotate forward:** draw the older item in that place toward the front of that place, so it stands in front of everything that goes in after it. It is done before any new item goes into that place.
- **Behind:** further from the front of that place. On the shelf, behind means nearer the back of the shelf. In the drawer, behind means nearer the back of the drawer. In a bin, behind means nearer the door face.
- **Front approach:** the gripper enters and leaves the door mouth level and from the front. It never comes down onto the working shelf from above. Every action on the working shelf is a front approach.

- **Group:** the part of the working shelf where one kind of older stock already stands. A new item joins the group of its own kind: carton to carton, jar to jar, tub to tub.
- **Outer rail:** the front wall of a door bin, the one away from the door face. An older bin item is drawn forward until it stands against the outer rail.
- **Divider:** the fixed wall down the middle of the stock tote. It splits the tote into the left gripper's half and the right gripper's half. Nothing crosses it.
- **Stands on its own:** after the gripper moves clear, the item stays upright for 2 seconds and does not rock, lean, slide, or fall.
- **Upright and square:** the item stands on its own base with its label facing the front, not leaning, not lying on its side, and not turned to face a side wall.
- **Clear space:** two items in the same place do not touch each other, and the gap between them is visible from the front.
- **Shut flush:** the drawer face sits level with the front of the appliance, with no part of it standing out.
- **Sealed:** the door meets the cabinet all round, no gap shows anywhere along the seal, the latch holds, and the door does not spring back when the **right gripper** comes off.
- **Overhang the front edge:** any part of an item sticks out past the shelf front edge into the door mouth.

Steps

Step 1: Load the crisper drawer

Goal: the drawer is shut flush, with the older produce pack at its front and both new packs flat behind it.

The drawer is worked by the **right gripper** alone. The **left arm stays at home** for the whole of this step. The door stands open on its catch and nothing touches it.

1.1 Check the door stands open

- Read the door from the front. It stands open on its **door catch**, the whole door mouth is clear, and both bins face into the workspace on the left.
- Touch nothing. Neither gripper takes hold of the door.

Check: the door stands open on its own and does not drift. If it drifts or swings, do not work it. End the episode and report it.

1.2 Pull the drawer open

- With the **right gripper**, close on the crisper drawer handle.
- Draw the drawer straight out toward the front until it meets its stop. Do not jerk it. Do not pull after it has stopped.

- Do not lift the drawer and do not lean on the drawer face.
- Open the **right gripper** and draw it clear of the drawer.

Check: the drawer stands open at its stop and the whole drawer floor is visible. If the drawer will not run out, do not force it. Leave it for the station.

1.3 Rotate the older produce forward

- With the **right gripper** closed, come down beside the older produce pack and set the gripper against its back edge.
- Push the pack forward along the drawer floor until it lies against the drawer front wall.
- Keep it flat the whole way. Do not lift it. Do not take it out of the drawer.

Check: the older pack lies flat against the drawer front wall, with clear drawer floor behind it. If it turned as it slid, square it with the closed **right gripper** against its edges.

1.4 Lay the two new packs in behind it

- With the **right gripper**, lift one produce pack straight up out of the **right half** of the stock tote. Take an item that has nothing on top of it. Never dig under another item.
- Carry it level to the drawer and lay it flat on the drawer floor behind the older pack, with clear space between them.
- Open the **right gripper** and draw it straight up clear of the drawer.
- Do the same with the second produce pack, laying it flat behind the first new pack.
- Lay every pack flat. Never stack one pack on another. Never stand a pack on edge.

Check: three packs lie flat in one row, older pack at the front and both new packs behind it, none stacked and none standing above the drawer rim. If a pack sits on top of another, lift it with the **right gripper** and lay it flat behind.

1.5 Push the drawer shut

- With the **right gripper** closed, press the drawer face flat and push it straight back until it is shut flush.
- Push until the drawer stops. Do not push hard enough to move the appliance.
- Draw the **right gripper** back clear of the door mouth.

Check: the drawer is shut flush, nothing is caught in it, and the base has not moved. If the drawer stops short, draw it open with the **right gripper**, lay the packs flat again, and shut it once more.

Expected state: door open on its catch, drawer shut with three packs in it, working shelf still holding the three older items only, five new items left in the tote.

Step 2: Load the three shelf groups

Goal: each group holds its older item at the front and its new item behind it, all standing upright and square, with nothing overhanging the shelf front edge.

Work the groups in this fixed order: **carton group, then jar group, then tub group**. That is left to right, and it is the far group first for each arm, so no gripper ever reaches over an item it has just stood down.

The **left gripper** works the carton group. The **right gripper** works the jar group and the tub group. Neither arm goes into the other's lane. Only one arm moves at a time.

Run Steps 2.1 and 2.2 once per group, three times.

2.1 Rotate the older item forward

- With the gripper for that group closed, enter the door mouth from the front, level, and set the gripper against the back of the older item of that group.
- Draw the item forward along the shelf until it stands just inside the shelf front edge.
- Keep it standing the whole way. Do not lift it. Do not tip it. Do not take it out of the refrigerator.
- Draw the gripper straight out of the door mouth.

Check: the older item stands upright and square just inside the shelf front edge, nothing sticks out past that edge, and there is clear shelf behind it. If it leans, stand it up with the same gripper. If it stands out past the edge, press it back with that gripper closed.

2.2 Stand the new item in behind it

- With the gripper for that group, close on the item of that kind in its own half of the stock tote: the **left gripper** on the carton in the left half, the **right gripper** on the jar and then the tub in the right half. Grip around the item's body, not by its lid or cap.
- Lift it straight up out of the tote, carry it level to the door mouth, and take it in from the front. Do not carry it over items already standing on the shelf. Do not carry it over the door or the bins. Do not carry it across the middle of the base.
- Stand it down on the shelf behind the older item of its group, upright and square, its label facing the front, with clear space between the two.
- Open the gripper and draw it straight out of the door mouth.

Check after each group: two items of that kind stand in the group, older at the front and new behind, both upright and square, both labels facing the front, clear space between them, and nothing overhanging the shelf front edge.

If the new item leans or touches the older one, take it back up with the same gripper and stand it down again. If it went to the wrong group, take it up with the gripper that owns the group it landed in and stand it with the older item of its own kind.

Expected state after the third group: six items on the working shelf, two of each kind, older in front of new in all three groups. The bottle and the small jar are left in the tote's left half.

Step 3: Load the two door bins

Goal: each door bin holds its older item against the outer rail with its new item behind it, and both items stand on their own.

Both bins sit on the open door in the **left lane**. So the **left gripper** does this whole step. The **right arm stays at home**. The door stays on its catch and is not touched.

Work the **tall bin first, then the short bin**, so the **left gripper** never reaches over an item it has just stood down.

3.1 Load the tall bin

- With the **left gripper** closed, come straight down into the tall bin and set the gripper against the back of the older bottle.
- Draw the bottle forward until it stands against the outer rail. Keep it standing. Do not lift it out.
- Draw the **left gripper** straight up out of the bin.
- With the **left gripper**, close on the new bottle in the **left half** of the stock tote, around its body. Lift it straight up and carry it level to the bin.
- Lower it straight down into the bin behind the older bottle. Stand it on the bin floor. Only then open the **left gripper**. Never drop or throw an item into a bin.
- Draw the **left gripper** straight up clear of the bin.

Check: both bottles stand upright in the tall bin, older against the outer rail and new behind it, clear space between them, and nothing standing high enough to catch the shelves when the door swings. The door has not moved on its catch.

3.2 Load the short bin

- Run the same motions as Step 3.1 with the **left gripper**, using the older small jar and the new small jar in the short bin. Draw the older jar forward to the outer rail, then stand the new jar down behind it.

Check: both jars stand upright in the short bin, older against the outer rail and new behind it, with clear space between them. The stock tote is empty on both sides.

Expected state: all seven new items away, the tote empty, the drawer shut, and the door still open on its catch.

Step 4: Close the door fully

Goal: the door is shut and sealed, with no gap along the seal and no spring back.

The door swings from the left, so the **left gripper** pushes it round. It seals at its free edge on the right, so the **right gripper** presses it home. Neither arm crosses the other.

4.1 Push the door round

- Check first that both grippers are out of the door mouth and empty, that the drawer is shut, and that nothing overhangs the shelf front edge.
- With the **left gripper** closed, set it on the door's **inner face low down near the hinge**, clear of both bins.
- Push steadily until the door comes off its **door catch**, then keep pushing it round toward the door mouth in one smooth move. Do not push fast. Do not let the door hit the appliance.
- Stop when the door has come round in front of the door mouth and rests against the cabinet.
- Draw the **left gripper** straight back off the door and keep it clear until the episode ends.

Check: the door hangs in front of the door mouth, nothing is caught between the door and the appliance, and nothing has been knocked over inside. If an item was knocked over, push the door open onto its catch again with the **left gripper**, stand the item up with the gripper that owns that place, and push the door round once more.

4.2 Press the door home

- With the **right gripper** closed, press the door's front face flat, at the **free edge side on the right**, straight back toward the appliance.
- Press until the door meets the cabinet all round and the latch holds. Hold the press for 2 seconds.
- Press straight back only. Do not slam the door. Do not press at an angle or on the door edge.
- Draw the **right gripper** straight back off the door.

Check: the door is **sealed**. It meets the cabinet all round, no gap shows anywhere along the seal, and the door does not spring back when the **right gripper** comes off.

If a gap shows or the door springs back, press it home once more with the **right gripper**. If it still will not seal, push the door open onto its catch with the **left gripper** and look for an item overhanging the shelf front edge, a pack caught in the drawer, or an item standing proud of a bin.

Expected state: door shut and sealed. Drawer shut flush. Tote empty on its stand. Base still locked in its park.

Step 5: End the episode

- Confirm the door. It is shut and sealed, with no gap along the seal and no spring back.
- Confirm the drawer. It is shut flush, with nothing caught in it.

- Confirm the stand. The stock tote is empty on both sides and standing where it started.
- Confirm nothing has been left on the stand or dropped on the floor.
- Confirm the base has not moved and the appliance has not shifted.
- Correct any failed check before ending.
- Return both arms home with grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Open the door by hand and set it on its catch, wide enough that the whole door mouth is clear.
2. Take the new carton, jar, and tub off the working shelf. Stand the carton in the tote's left half and the jar and the tub in its right half, upright and not touching.
3. Slide each older carton, jar, and tub back to the back of its own group, upright and square, label facing the front, with the three groups clear of one another.
4. Take the new bottle out of the tall bin and the new small jar out of the short bin. Stand both in the tote's left half.
5. Slide the older bottle and the older small jar back against the door face at the back of their own bins.
6. Pull the drawer open, take the two new produce packs out, and stand them in the tote's right half.
7. Push the older produce pack back to the back of the drawer, lying flat, and shut the drawer flush.
8. Check the tote holds all seven new items standing upright, three in the left half and four in the right half, none on top of another and none touching, and none on the wrong side of the divider.
9. Wipe up any spill on the shelf, in a bin, in the drawer, or on the stand, and dry it.
10. Replace an item that leaks, that no longer stands on its own base, or whose lid or cap has come loose. Replace a produce pack that has split or lost its shape.
11. Check the door still latches and seals when it is pushed shut by hand, that it does not spring back, and that its catch still holds it open on its own.
12. Check the drawer still runs to its stop and shuts flush.
13. Check the stock stand is still steady and does not slide when an item is lifted out of the tote.
14. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the refrigerator shifts in frame, the door mouth changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 5. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Arms crossed

- **Visible cue:** the **left gripper** touches the drawer, the jar group, the tub group, or the right half of the tote. The **right gripper** touches the carton group, a door bin, the door catch, or the left half of the tote. The left gripper goes to the right of the right gripper. An item is taken across the tote's divider. One arm reaches over, under, around, or past the other, or across the front of the other arm's body.
- **SOP rule broken:** Arm lanes and Arm assignments. The left gripper works the left lane, the right gripper works the right lane, and the left gripper always stays left of the right gripper.
- **Coaching note:** the divider tells you whose item it is. If you have to reach past the other arm, the wrong arm is doing the job.

Violation: Door held or moved during loading

- **Visible cue:** a gripper takes hold of the door, leans on it, or pushes it during Steps 1 to 3. The door comes off its catch before Step 4. Or the door drifts in or out while the shelf or the bins are worked.
- **SOP rule broken:** Steps 1.1 to 3.2. The **door catch** holds the door open, and neither gripper touches the door until Step 4.
- **Coaching note:** the catch holds the door. Both arms are for stock until the loading is done.

Violation: Approached the shelf from above or touched the shelf above

- **Visible cue:** a gripper comes down onto the working shelf from above instead of entering level from the front. Or a wrist or forearm knocks, scrapes, or rests on the shelf above or the shelf front edge on the way in or out.
- **SOP rule broken:** Steps 1 to 3. Every action on the working shelf is a front approach, level, in and out through the door mouth.
- **Coaching note:** straight in from the front, straight out the same way. There is no way in from the top.

Violation: Leaned on or pushed the appliance

- **Visible cue:** a gripper, wrist, or forearm rests on the door, the drawer face, or the shelf edge. The appliance rocks or shifts. Or a push on the drawer or the door moves the whole appliance.
- **SOP rule broken:** Steps 1 to 4. The refrigerator and the door carry no weight and are never pushed out of place.
- **Coaching note:** the arm holds itself up. Press only as hard as the drawer or the latch needs.

Violation: Wrong loading order

- **Visible cue:** the shelf or a bin is loaded before the drawer is shut. The bins are loaded before all three groups are done. The groups are not worked carton, jar, tub. The short bin is loaded before the tall bin. Or a place is skipped and come back to.
- **SOP rule broken:** Steps 1 to 3. Load the drawer and shut it, then the three groups left to right, then the tall bin and the short bin.
- **Coaching note:** drawer, shelf, bins, door. Far side first every time, so nothing is carried over what has just been stood down.

Violation: Drawer opened or shut wrong

- **Visible cue:** the **right gripper** jerks the drawer, keeps pulling after it has met its stop, lifts the drawer, forces a drawer that will not run, or shuts it without the face sitting flush.
- **SOP rule broken:** Steps 1.2 and 1.5. Draw the drawer straight out to its stop and push it straight back until it is shut flush.
- **Coaching note:** straight out to the stop, straight back to flush, and let go of the handle before the push.

Violation: Older stock not rotated forward

- **Visible cue:** a new item goes into a group, a bin, or the drawer while the older item there is still at the back. Or the older item is only part way forward and is left there.
- **SOP rule broken:** Steps 1.3, 2.1, 3.1, and 3.2. Draw the older item forward to the front of its place before any new item goes in behind it.

- **Coaching note:** rotate first, then fill. Every place gets the old one moved before the new one arrives.

Violation: New stock put in front of older stock

- **Visible cue:** a new item ends up nearer the front of a group, a bin, or the drawer than the older item of that place.
- **SOP rule broken:** Steps 1.4, 2.2, 3.1, and 3.2. The new item is stood down behind the older item every time.
- **Coaching note:** the older item is always the one you see first from the front. If the new one is in front, it is wrong.

Violation: Older stock lifted or taken out

- **Visible cue:** an older item is lifted off the shelf, out of a bin, or out of the drawer. It is carried in the air, or set down outside the refrigerator.
- **SOP rule broken:** Steps 1.3, 2.1, 3.1, and 3.2. Older stock is slid along the surface it stands on and never leaves the refrigerator.
- **Coaching note:** slide it, do not lift it. Nothing that starts inside comes out.

Violation: Produce stacked or laid wrong

- **Visible cue:** a produce pack is laid on top of another, stood on edge, laid across the row instead of along it, or left standing above the drawer rim.
- **SOP rule broken:** Step 1.4. Lay every pack flat on the drawer floor in one row, none stacked.
- **Coaching note:** flat, in the row, one behind the other. A stacked pack blocks the drawer from shutting.

Violation: Item put with the wrong group

- **Visible cue:** a gripper stands a carton with the jars, a jar with the tubs, or any new item anywhere other than with the older item of its own kind. Or a bottle or small jar is stood on the shelf.
- **SOP rule broken:** Step 2.2. Each new item joins the group of its own kind: carton to carton, jar to jar, tub to tub.
- **Coaching note:** read what is already standing there before the arm goes in. Like joins like.

Violation: Item not stood upright and square

- **Visible cue:** an item on the shelf or in a bin leans, lies on its side, stands turned so its label faces a side wall or the back, touches its neighbour, or rocks after the gripper moves clear.
- **SOP rule broken:** Steps 2.2, 3.1, and 3.2. Stand every item on its own base, label facing the front, with clear space between the two items in that place.

- **Coaching note:** stand it down, watch it settle, then let go. Labels face the front so the next restock can read them.

Violation: Item left overhanging the shelf front edge

- **Visible cue:** an item on the working shelf sticks out past the shelf front edge into the door mouth when the gripper moves on. Or the door meets an item when it swings round.
- **SOP rule broken:** Steps 2.1 and 2.2. Every item stands inside the shelf front edge, with nothing sticking out past it.
- **Coaching note:** an item over the edge is a door that will not shut. Press it back before you move on.

Violation: Item taken from the tote wrong

- **Visible cue:** a gripper digs under an item, lifts two items together, drags an item up the tote wall, takes an item from the other half of the tote, reaches over the divider, or knocks or tips the tote or the stand.
- **SOP rule broken:** Steps 1.4, 2.2, 3.1, and 3.2. Take one item at a time, from your own half, with nothing on top of it, and lift it straight up out of the tote.
- **Coaching note:** one item, straight up, your own side of the divider, and the tote stays where it stands.

Violation: Item gripped by its lid or cap

- **Visible cue:** a gripper closes on a carton flap, a jar lid, a tub lid, or a bottle cap instead of around the item's body. Or a lid comes loose or lifts off as the item is carried.
- **SOP rule broken:** Steps 2.2, 3.1, and 3.2. Close around the item's body, not on its lid or cap.
- **Coaching note:** grip the body low, where the item is solid. A lid is not a handle.

Violation: Item carried over the loaded shelf, the bins, or the open drawer

- **Visible cue:** a gripper carries an item over items already standing on the shelf, over the door bins, or over the open drawer, instead of going in level through the door mouth or straight down into a bin.
- **SOP rule broken:** Steps 1.4, 2.2, 3.1, and 3.2. Carry each item level to the door mouth and take it in from the front, or lower it straight down into its bin.
- **Coaching note:** in from the front for the shelf, down from above for the bins, and never over something already placed.

Violation: Item dropped into a bin

- **Visible cue:** the **left gripper** opens above the bin and the item falls in. The item bounces or topples on landing. Or it is thrown at the bin.

- **SOP rule broken:** Steps 3.1 and 3.2. Lower the item until it stands on the bin floor, then open the gripper.
- **Coaching note:** stand it down, then let go. A dropped bottle swings the whole door.

Violation: Door pushed round with an arm still inside or still holding an item

- **Visible cue:** the **left gripper** pushes the door off its catch while either gripper is still inside the door mouth or still holding an item. Or the door catches an arm or an item as it comes round.
- **SOP rule broken:** Step 4.1. Check both grippers are out of the door mouth and empty before the door is pushed round.
- **Coaching note:** out and empty first, then push. The door does not wait for a hand.

Violation: Door pushed round by the wrong gripper or slammed

- **Visible cue:** the **right gripper** pushes the door round. The **left gripper** takes the door somewhere other than its inner face low near the hinge. The push is made over a bin. Or the door is swung fast enough to bang against the appliance.
- **SOP rule broken:** Step 4.1. The **left gripper** pushes on the door's inner face low near the hinge and brings it round in one smooth move.
- **Coaching note:** left gripper, hinge side, low and smooth. A slammed door topples what you just stood up.

Violation: Door not pressed fully home

- **Visible cue:** a gap shows along the seal after the **right gripper** comes off. The door springs back open. The latch does not hold. The press is released before 2 seconds. Or the press is made at an angle, on the door edge, on a bin, or by the **left gripper**.
- **SOP rule broken:** Step 4.2. The **right gripper** presses the door's front face flat and straight back at the free edge side, until it meets the cabinet all round and the latch holds, and holds the press for 2 seconds.
- **Coaching note:** flat face, straight back, hold two seconds, then watch it before you move away. A door left ajar is a failed restock.

Violation: New stock left in the tote

- **Visible cue:** the episode moves on to Step 4 with an item still standing in either half of the stock tote.
- **SOP rule broken:** Steps 1 to 3. All seven new items go into the refrigerator: two produce packs, three shelf items, two bin items.
- **Coaching note:** read both halves of the tote before the door moves. It is empty when the loading is done.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped. Or a check is made and the fault it finds is left uncorrected: a leaning item, a wrong group, an item over the front edge, a stacked pack, a drawer not flush, or a door showing a gap.
- **SOP rule broken:** Steps 1.1 to 4.2. Run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, toppled, or knocked over

- **Visible cue:** an item is dropped on the stand or the floor. An item already standing is toppled by an arm or by the door. The drawer contents shift when it shuts. Or the tote or the stand is knocked out of place.
- **SOP rule broken:** Steps 1 to 4. Nothing is dropped, toppled, or knocked out of its place, and every gripper leaves the door mouth straight out to the front or straight up.
- **Coaching note:** check the path and the landing place before the arm moves, and come out the way you went in.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** works the drawer, the jar group, or the tub group, or presses the door home. The **right gripper** works the carton group or a door bin, or pushes the door off its catch. Or an item is handed from one gripper to the other.
- **SOP rule broken:** Steps 1 to 4. The **left gripper** works the carton group, both bins, and the door push. The **right gripper** works the drawer, the jar group, the tub group, and the final press. Nothing changes hands.
- **Coaching note:** left arm left, right arm right, and nothing changes hands.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the door open or showing a gap, an item still in the tote, the drawer standing open, an item leaning or with the wrong group, the tote or the stand knocked out of place, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 5. Confirm the door, the drawer, the stand, the floor, and the base. Then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).

- **Camera cannot read the shelf groups, a bin, or the door seal**, so where an item went or whether the door shut cannot be judged.
- **Hardware fault on an arm**: gripper failure, drift, controller caused collision, or motor error.
- **Base fault**: a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Faulty door catch**: a catch that will not hold the door open on its own, that lets the door drift, or that will not release under a correct steady push.
- **Faulty door**: a door whose latch will not hold, whose seal will not close under a correct press, or that swings shut on its own.
- **Faulty drawer**: a drawer that jams, that will not run to its stop, that comes free of the appliance, or that will not shut flush when correctly loaded.
- **Door mouth too tight**: a mouth so low or so narrow that a gripper cannot enter level from the front, or cannot reach the back of the working shelf.
- **Shelf will not hold an item**: a shelf that is not level, or so slick that a correctly stood item will not stay where it is put.
- **Appliance not steady**: a refrigerator that rocks or walks when the drawer is drawn or the door is pressed shut.
- **Defective item**: one that leaks, that will not stand on its own base, whose lid or cap comes off in a correct grip, or that slips out of a correct grip.
- **Defective produce pack**: one that has split, lost its shape, or will not lie flat on the drawer floor.
- **Group has no room**: a stretch of shelf that will not hold its two items standing one behind the other with clear space between them.
- **Bin too small**: a bin that will not hold its two items standing one behind the other, or that is too shallow for the bottle to stand in it.
- **Stock stand will not stay put**: a stand or tote that slides or tips when an item is lifted straight up out of it.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so a half of the tote, the drawer, an end of the working shelf, or a bin cannot be reached without extending or folding the arm, or without reaching into the other arm's lane.

Annotation subtasks (from SOP)

1. Pull the crisper drawer open
2. Slide the older produce pack forward in the drawer
3. Lay one new produce pack flat in the drawer
4. Push the crisper drawer shut
5. Slide the older item forward in one shelf group

6. Lift one item out of its half of the stock tote
7. Stand one new item down behind the older item in its group
8. Slide the older bin item forward to the outer rail
9. Stand one new item down in a door bin
10. Push the door off its catch and round to the door mouth
11. Press the door home until it seals
12. Return both arms home and end the episode